Ex-6

Step-0 Libraries Used

Pandas: is used to load and handle the dataset.
NumPy: helps in numerical operations.
Matplotlib and Seaborn: are used to visualize class distribution and model comparison graphs.
Scikit-learn: is used for train-test splitting, label encoding, model training, and evaluation. LogisticRegression, RandomForestClassifier, and XGBClassifier are used as classification models.
The metrics module: is used to calculate Accuracy, Precision, Recall, F1 (weighted, macro, micro), ROC-AUC, and Confusion Matrix.
Pickle: is used to save the trained best model for future use without retraining.

Step-1 Problem Definition

The objective of this experiment is to build a credit scoring model that classifies loan applicants into risk categories based on historical financial information.Since the output variable Credit_Risk contains multiple categories (High, Medium, Low), this is a multiclass classification problem.The goal is to compare different classification models and select the best-performing one.

Step-2 Data Understanding

The dataset contains 500 records and 11 columns
Important columns include:

- Age
- Annual_Income
- Employment_Years
- Credit_Score
- Loan_Amount
- Loan_Term_Months
- Existing_Loans_Count
- Debt_to_Income_Ratio
- Late_Payments_Last_2Yrs
- Credit_Risk (Target Variable)

The target column Credit_Risk is categorical (object type)
A countplot was generated to visualize the distribution of risk classes This helps understand class imbalance before training models.

Step-3 Data Processing

The target column Credit_Risk was converted into numerical format using LabelEncoder, because machine learning algorithms require numerical input.The Applicant_ID column was dropped since it is only an identifier and does not contribute to prediction.Features (X) were separated from the target (y).

Step-4 Train-Test Split and Model Training

The dataset was split using:
train_test_split(test_size=0.2, random_state=0)

This means:80% data used for training,20% data used for testing

Three models were trained:

- **Logistic Regression:**Linear model for classification.Performs well when classes are linearly separable.
- **Random Forest:**Ensemble of multiple decision trees.Reduces overfitting and improves stability.
- **XGBoost:**Gradient boosting algorithm. Builds trees sequentially to correct previous errors. Known for high performance in structured data.

Step-5 Model Evaluation and Comparative Analysis

Each model was evaluated using:

- Accuracy
- Precision (weighted)
- Recall (weighted)
- F1_weighted
- F1_macro
- F1_micro

Model Accuracy:

- Logistic Regression=0.68
- Random Forest=0.99
- XGBoost=1.00

XGBoost achieved perfect classification performance.

Step-6 Prediction

The confusion matrix shows:

- Class 0 = 70 correctly predicted
- Class 1 = 4 correctly predicted
- Class 2 = 26 correctly predicted

No misclassifications occurred.

The classification report confirms precision, recall, and F1-score of 1.00 for all classes
This indicates perfect separation of risk categories.

Step-7 Best Model Selection and Model Saving

The best model was selected based on highest F1_weighted score:
Best Model: XGBoost
The trained best model was saved using pickle:
pickle.dump(best_model,f)

Saving the model ensures:

- No need to retrain
- Reproducibility
- Deployment readiness

The saved model can later be loaded and used for predictions.